

Q1

A logistics company needs to deliver packages to multiple destinations from a central warehouse. The goal is to determine the most efficient route that minimizes travel time or distance while ensuring all destinations are visited.

Propose a suitable algorithm to solve the problem. (Correct answer: 2p)

Skriv ditt svar her

Could be Dijkstra algorithm to find the shortest path



Q2

What is the purpose of heaps? (2p)

Skriv ditt svar her

The primary purpose of heaps is to efficiently manage and access the smallest or largest element in a dataset, depending on the type of heap used. Heaps are particularly useful in applications that require priority management or sorting.



Q3

In a level-order traversal of a binary tree, which data structure is commonly used? (2p)

Velg ett alternativ:

☐ Linked List

☒ Queue



☐ Stack

☐ Heap

Q4

What is the time complexity of inserting an element into AVL tree in the worst case? (2p)

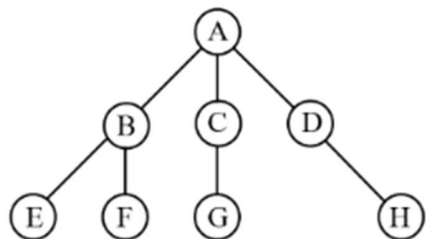
Velg ett alternativ:

- ☐ $O(1)$
- ☐ $O(n)$
- ☐ $O(n \log n)$
- ☒ $O(\log n)$



Q5

Who is the successor of node G? (2p)



Velg ett alternativ:

- ☐ A
- ☐ F
- ☒ C
- ☐ H



Q6

AVL trees have a lookup time roughly equal to that of which other data structure? (2p)

Velg ett alternativ:

- ☐ Hash Table
- ☐ Unsorted Array
- ☐ None of the Above
- ☒ Sorted Array
- ☐ Linked List



Q7

What is the time complexity of the given function? (2p)

```
a = 0
b = 0
for i in range(N):
    a = a + random()

for i in range(M):
    b = b + random()
```

Velg ett alternativ:

- ☐ $O(N)$
- ☐ $O(N \cdot M)$
- ☒ $O(N+M)$
- ☐ $O(n+m)$



Q8

what is the time complexity of the given function? (2 p)

•

```
k = 0;
for i in range(n//2,n):
    for j in range(2,n,pow(2,j)):
        k = k + n / 2;
```

Velg ett alternativ:

☒ $O(n \log n)$



☐ $O(\log n)$

☐ $O(n^2 \log n)$

☐ $O(n)$

Q9

what is the time complexity of the given function? (2 p)

```
def fun(n,m):
    arr=[[0]*m for i in range(n)]
    for i in range(n):
        for j in range(m):
            k=1
            while k<n*m:
                k*=2
```

Velg ett alternativ:

☐ $O(n*m)$

☐ $O(n)$

☒ $O(n*m \log(n*m))$



☐ $O(n*m \log(n))$

Q10

Which of the given options provides the increasing order of asymptotic complexity of functions f_1 , f_2 , f_3 and f_4 ? (2p)

A. $f_2(n) = n^{3/2}$

B. $f_1(n) = 2^n$

C. $f_4(n) = n^{\log n}$

D. $f_3(n) = n \log n$

Choose the correct answer.

Velg ett alternativ:

☐ B, C, D, A

☐ B, C, A, D

☒ D, A, C, B



☐ C, B, A, D

Q11

Consider an array which stores a maximum of 100 elements. For case 1, the user inserts 1 element into an array. For case 2, the user inserts 100 elements into an array. What is the time complexity for both the cases? (2p)

Velg ett alternativ:

☐ $O(1)$, $O(n)$

☐ $O(n)$, $O(1)$

☒ $O(1)$, $O(1)$



☐ $O(n)$, $O(n)$

Q12

in the given recursive function, what would be the result of function with the given input.
 $n = 3$.

(4 p)

```
def f(n):  
    if n == 0 : return 0  
    if n == 1 : return 1  
    else:  
        return f(n-1) + f(n-2) + 1
```

Velg ett alternativ:

☒ 4



☐ 15

☐ 12

☐ 8

Q13

What is the space complexity needed to store a linked list of n nodes? (2p)

Velg ett alternativ:

☐ None of them

☐ $O(1)$

☒ $O(n)$



☐ $O(n^2)$

Q14

Which type of linked list stores the address of the head node in the next pointer of the last node? (2p)

Velg ett alternativ:

- ☐ Hashed List
- ☐ Singular Linked List
- ☒ Circular Linked List
- ☐ Doubly Linked List



Q15

Consider a sequence of push and pop operations used to push the integers 0 through 9 on a stack. The numbers will be pushed in order, however the pop operations can be interleaved with the push operations, and can occur any time there is at least one item on the stack. When an item is popped, it is printed to the terminal.

Which of the following could NOT be the output from such a sequence of operations? (4p)

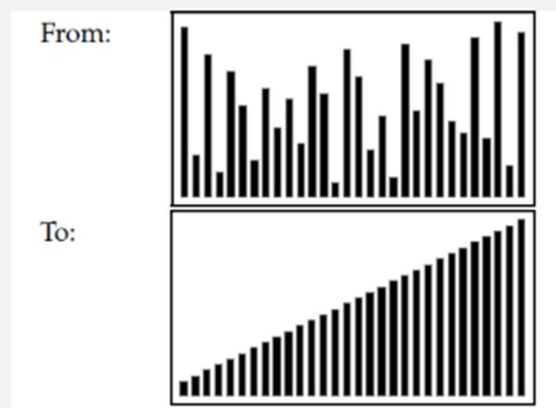
Velg ett alternativ:

- ☒ 5 6 7 8 9 0 1 2 3 4
- ☐ 0 1 2 3 4 5 6 7 8 9
- ☐ 4 3 2 1 0 5 6 7 8 9
- ☐ 4 3 2 1 0 9 8 7 6 5

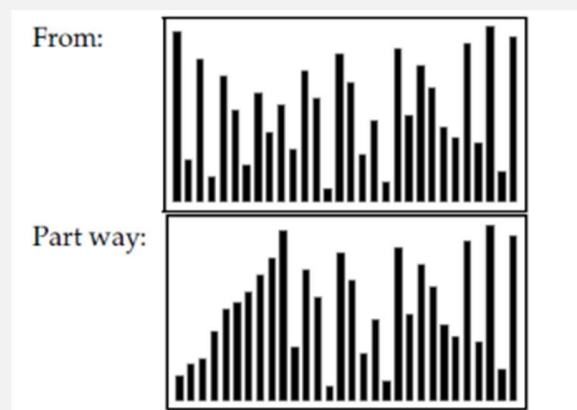


Q16

Suppose we want to sort a collection of sticks according to height:



Each of the following diagrams show the state of the sticks part way through sorting using a different sorting algorithm. For each diagram, say which algorithm is being used. (2p)



Velg ett alternativ:

☒ insertion sort



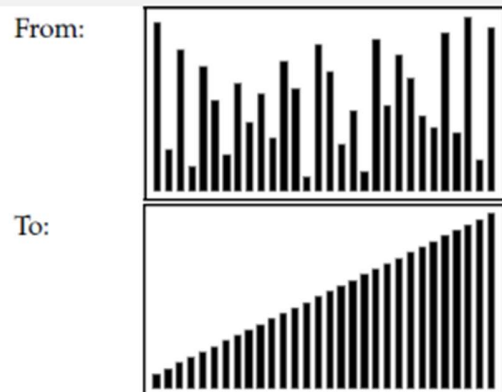
☐ selection sort

☐ bubble sort

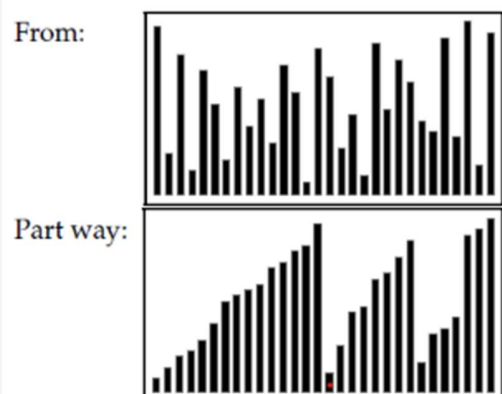
☐ merge sort

Q17

Suppose we want to sort a collection of sticks according to height:



Each of the following diagrams show the state of the sticks part way through sorting using a different sorting algorithm. For each diagram, say which algorithm is being used. (2p)



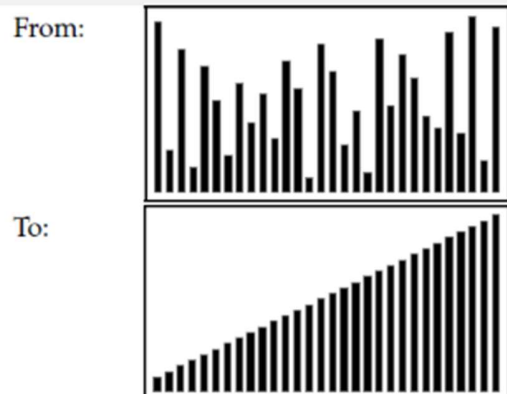
Velg ett alternativ:

- ☐ insertion sort,
- ☒ mergeSort
- ☐ selection sort
- ☐ bubble sort

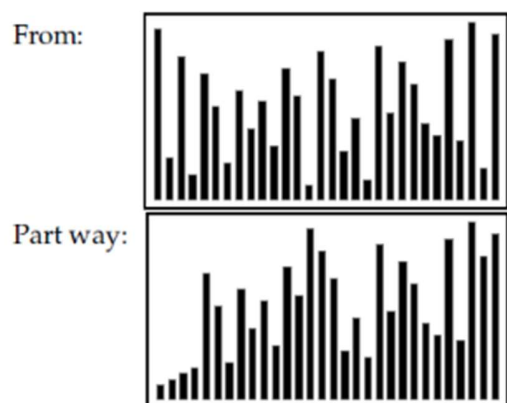


Q18

Suppose we want to sort a collection of sticks according to height:



Each of the following diagrams show the state of the sticks part way through sorting using a different sorting algorithm. For each diagram, say which algorithm is being used. (2p)



Velg ett alternativ:

- ☐ insertion sort
- ☒ selection sort
- ☐ merge sort
- ☐ bubble sort



Q19

Which of the following represents the sequence of operations to reverse a string using a stack?

Velg ett alternativ:

- ☐ Enqueue all characters, then dequeue in reverse order
- ☒ Push all characters onto the stack, then pop characters in reverse order
- ☐ None of the above
- ☐ Push all characters, then push them again in reverse order



Q20

Given an array `arr` and an integer `k`, what is the goal of the task??
Write the output of the function. (2p)

```
1  # Python program to find k largest elements in an
2  # array using sorting
3
4  def kLargest(arr, k):
5      arr.sort(reverse=True)
6      res = arr[:k]
7      return res
8
9  arr = [1, 23, 12, 9, 30, 2, 50]
10 k = 3
11 res = kLargest(arr, k)
12 print(res)
13
```

[50, 30, 23]

////////////////////////////////////

The given function is going to find the sum of the values of the nodes in the singly linked list. Fill in the blank.

```
def sumOfNodes(head):  
    temp = head  
    sum = 0  
    while (-----):  
        -----  
        -----  
    return sum
```

Solution:

```
def sumOfNodes(head) :  
    temp = head  
    sum = 0  
    while (temp != None) :  
        sum = sum + temp.data  
        temp = temp.next  
    return sum
```

////////////////////

Trace the **Quicksort algorithm** over the array here. A = [45,23,87,12,72,4,54,32,52]

What would be the array after the first complete iteration? Take 45 as pivot.

A = [4,23,32,12,45,72,54,87,52]

////////////////////////////////////

Suppose you have the following hash table, implemented using *linear probing*. The hash function we are using is the identity function, $h(x) = x$.

0	1	2	3	4	5	6	7	8
9	18		12	3	14	4	21	

In which order could the elements have been added to the hash table?
There are several correct answers, and you should give all of them.
Assume that the hash table has never been resized, and no elements have been deleted yet.

- A 9, 14, 4, 18, 12, 3, 21
- B 12, 3, 14, 18, 4, 9, 21
- C 12, 14, 3, 9, 4, 18, 21
- D 9, 12, 14, 3, 4, 21, 18
- E 12, 9, 18, 3, 14, 21, 4

Answer: C and D

In A, 4 would be inserted at index 4 instead of 6
In B, 18 would be inserted at index 0 instead of 1
In E, 21 would be inserted at index 6 instead of 7

For VG only: if we want a hash table that stores a set of *strings*, one possible hash function is the string's length, $h(x) = x.length$.

Is this a good hash function? Explain your answer.

Answer: no. Strings with the same length will have the same hash code. If we insert lots of strings with the same length, lookup will take $O(n)$ time instead of $O(1)$.

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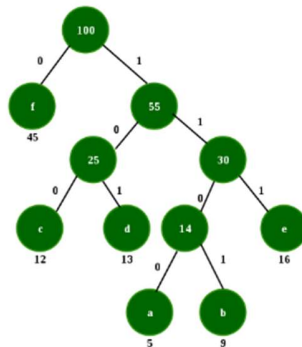
Huffman Tree

A networking company uses a compression technique to encode the message before transmitting over the network. Suppose the message contains the following characters with their frequency:

Character	Frequency
a	5
b	9
c	12
d	13
e	16
f	45

Note that each character in input message takes **1 byte**. If the compression technique used is Huffman Coding, **how many bits will be saved in the message?** (8p)

Finding number of bits without using Huffman, Total number of characters = sum of frequencies = 100 size of 1 character = 1byte = 8 bits Total number of bits = $8 \times 100 = 800$ Using Huffman Encoding, Total number of bits needed can be calculated as: $5 \times 4 + 9 \times 4 + 12 \times 3 + 13 \times 3 + 16 \times 3 + 45 \times 1 = 224$ Bits saved = $800 - 224 = 576$.



- A 1100
- B 1101
- C 100
- D 101
- E 111
- F 0

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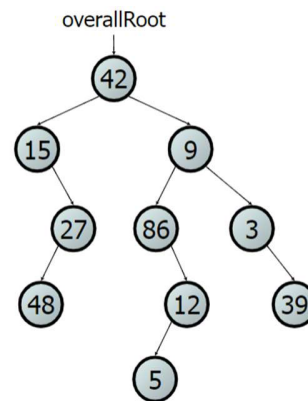
Reconstructing the tree based on the given pre-order and in-order traversal.

Find the post order traversal for the tree later on.

Pre: 42,15,27,48,9,86,12,5,3,39

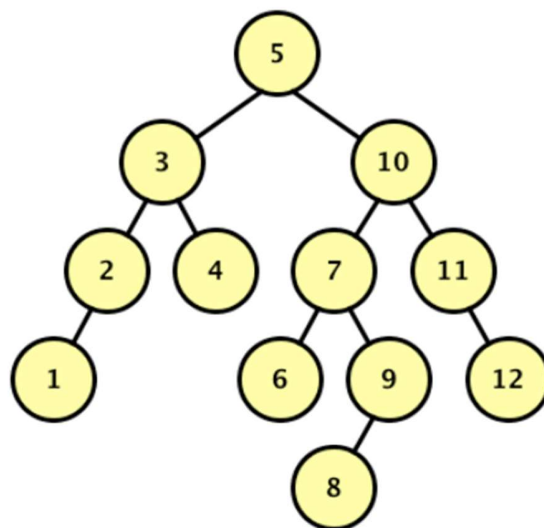
in: 15,48,27,42,86,5,12,9,3,39

Post order = 48, 27, 15, 5, 12, 86, 39, 3, 9, 42



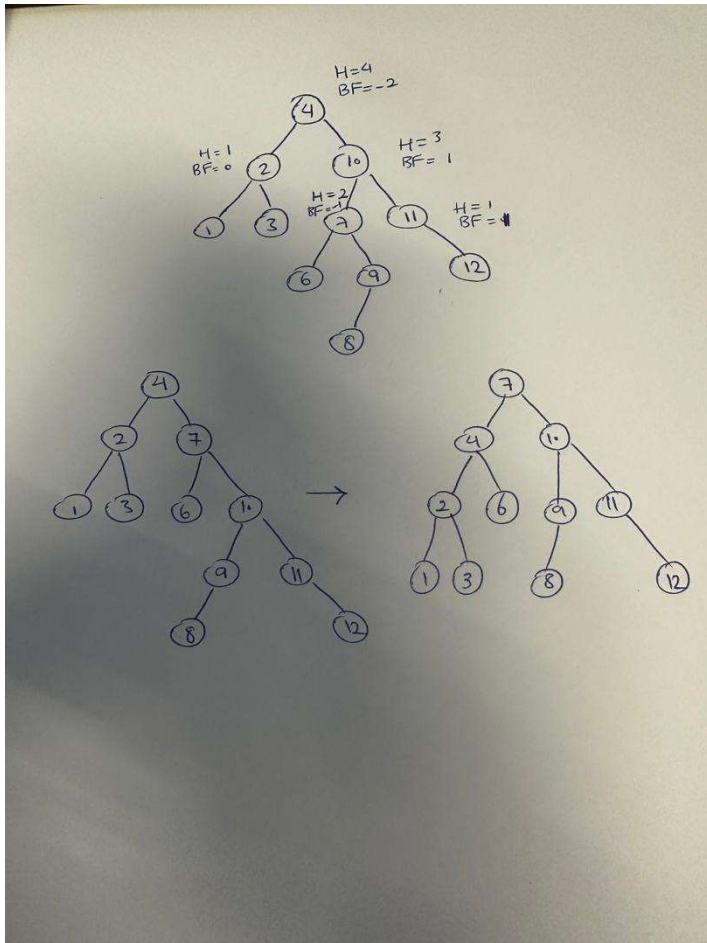
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Given the following AVL Tree:



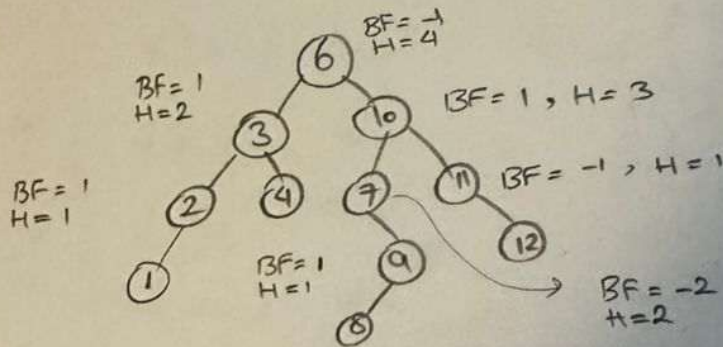
- a) Draw the resulting **BST** after **5 is removed**, but before any rebalancing takes place. Label each node in the resulting tree with its balance factor. (4p)
Use paper and draw the tree on the sheet.

- b) Now rebalance the tree that results from (a). Draw a new tree for each rotation that occurs when rebalancing the AVL Tree. You do not need to label these trees with balance factors. (6p)

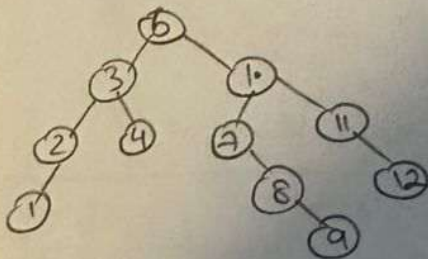


Or

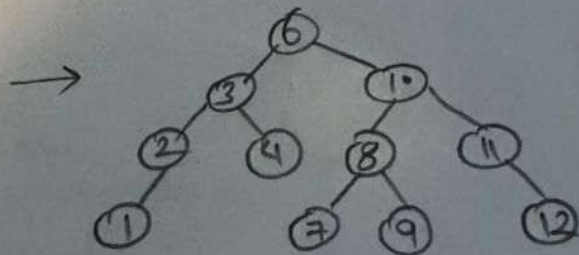
5 will be removed and replaced by the min value from the right subtree, which is 6. 6 is also Successor of 5.



Right rotation



Left rotation

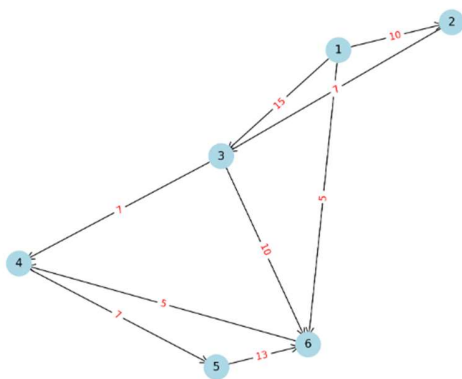


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a) Draw the graph corresponding to the following list of edges. **You should draw the graph on the sheet.** (4p)

from	to	cost
1	2	10
1	3	15
1	6	5
2	3	7
3	4	7
3	6	10
4	5	7
6	4	5
5	6	13

Solution:



The Breadth-First Search (BFS) traversal starting from node 1 visits the nodes in the following order:

1 → 2 → 3 → 6 → 4 → 5 [-]

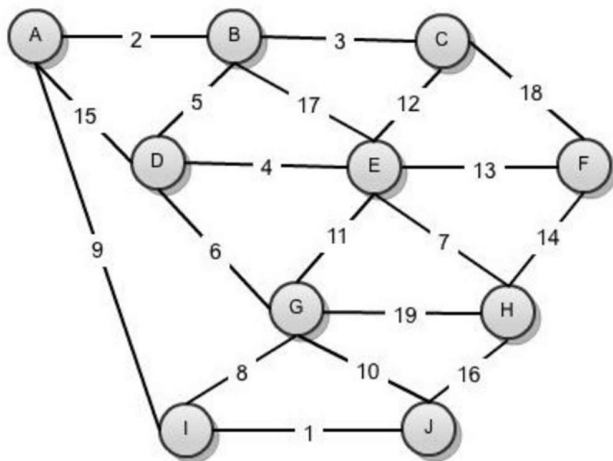
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Consider the graph you have drawn in the previous question. Show each step in applying **Dijkstra's algorithm** to the graph to find the shortest path starting from node 1 to other nodes. Please the shortest path clearly. (4p)

	2	3	4	5	6
1	10	15	10	17	5

//////////

Using **Prim's algorithm**, find the minimum weight spanning tree for the graph (6p)



Solution: $2 + 3 + 5 + 4 + 13 + 6 + 7 + 8 + 1 = 49$

